Lecture 18

Holonomy and Gauss–Bonnet

Objectives. *Parallel transport around a loop can rotate a tangent vector. On a surface, the rotation is the enclosed curvature, modulo a full turn. Keeping track of the boundary tangent and its corner turns gives the Gauss–Bonnet formula. Summing that formula over a closed surface relates total curvature to its Euler characteristic.*

By the end of this lecture you will be able to:

- Compute surface holonomy using a frame defined on the enclosed disk.
- Derive the angle excess and the boundary terms in local Gauss–Bonnet.
- Prove global Gauss–Bonnet given the cited topological inputs.
- Compute the latitude dependence of ideal Foucault precession.

18.1 Holonomy on an oriented surface

Let (M, g) be an oriented Riemannian surface with its Levi-Civita connection. Write $\mathrm{d}A$ for its metric area form. In a positive orthonormal frame $e = (e_1, e_2)$, recall Lecture 17's convention

$$\nabla e_1 = -\omega^1{}_2 e_2, \qquad \nabla e_2 = \omega^1{}_2 e_1, \qquad \mathrm{d}\omega^1{}_2 = K\mathrm{d}A.$$

The positive quarter-turn J is defined by $Je_1 = e_2$ and $Je_2 = -e_1$. It is independent of the positive orthonormal frame because plane rotations commute with this quarter-turn.

For a piecewise smooth loop γ based at p, its *holonomy* is the parallel-transport map $P_\gamma : T_pM \to T_pM$ of Lecture 16, composed along its smooth segments. It is a rotation. To see the angle and its sign, write a transported unit vector along a framed segment as

$$V = \cos\alpha\, e_1 + \sin\alpha\, e_2, \qquad \frac{DV}{dt} = (\dot\alpha - \omega^1{}_2(\dot\gamma))JV.$$

The derivative follows by the product rule and the two frame actions above. The angle α has a real lift along the interval by Lecture 3's circle lifting theorem. Parallel transport means $\dot\alpha = \omega^1{}_2(\dot\gamma)$. Thus its matrix between the positive frames at the segment endpoints is the rotation through $\alpha(b) - \alpha(a)$, measured counter-clockwise. The same construction works across finitely many curve segments. The figure illustrates the angle measured relative to a local frame.

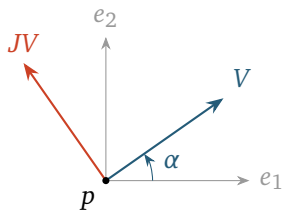


Figure 18.1: A unit vector has angle α from the first frame vector. Parallel transport changes this angle by the connection integral along the curve.

Theorem 18.1 (Surface holonomy · *proved given partitions of unity*)**.** *Let $D \subset M$ be a compact disk with smooth boundary, contained in an open set with one positive orthonormal frame. Parallel transport around ∂D, with its positive boundary orientation, rotates vectors by*

$$\Theta_D \equiv \int_D K \mathrm{d}A \pmod{2\pi}.$$

Proof. The frame agrees with itself when the loop returns to its starting point. Integrating the parallel-angle equation and using Stokes gives

$$\alpha(b) - \alpha(a) = \oint_{\partial D} \omega^1{}_2 = \int_D \mathrm{d}\omega^1{}_2 = \int_D K \mathrm{d}A.$$

A tangent-plane rotation records this real angle only modulo 2π. Linearity of transport extends the unit-vector conclusion to every vector. ∎

Reversing the loop negates the angle. A different positive frame along the loop changes the net lifted angle $\alpha(b) - \alpha(a)$ by an integer multiple of 2π. Indeed, its change-of-frame map is circle-valued, and its lifted angle has endpoint difference $2\pi m$ for some $m \in \mathbb{Z}$ by Theorem 3.7. The rotation is independent of that choice. Stokes requires a frame on the whole disk, not merely near its boundary.

18.2 Geodesic triangles and angle excess

A *geodesic triangle* bounds an embedded disk by three regular geodesic segments, with nondegenerate corners. Its interior angles α_1, α_2 and α_3 lie in $(0, 2\pi)$. The signed exterior turn at a vertex is $\varepsilon_i = \pi - \alpha_i \in (-\pi, \pi)$. It is the change from the incoming to the outgoing tangent, positive counter-clockwise in a positive tangent frame. The *angle excess* is $\alpha_1 + \alpha_2 + \alpha_3 - \pi$. The spherical triangle in the figure has positive excess.

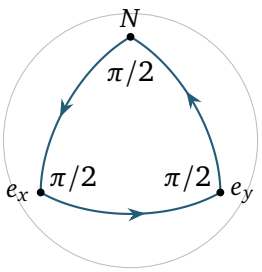


Figure 18.2: The three coordinate great-circle arcs bound an octant of the unit sphere. Each interior angle is $\pi/2$. Arrows give the positive boundary orientation, seen from outside.

We use the cited *turning-angle theorem*: for the positive boundary of a disk in one positive frame extending over the disk, the continuous changes of tangent angle along the sides plus the exterior turns add to 2π. See Tu, Theorem 17.4, pp. 142–143. A frame defined only on an annulus does not satisfy this hypothesis. For a proof in a chart, see Lee, Theorem 9.1 and Lemma 9.2, pp. 158–163.

Theorem 18.2 (Triangle excess · *proved given turning and partitions of unity*). *If a geodesic triangle D lies in one framed neighbourhood as above, then*

$$\alpha_1 + \alpha_2 + \alpha_3 - \pi = \int_D K \mathrm{d}A.$$

Proof. Along each side, the unit tangent is parallel, so its frame angle changes by the integral of $\omega^1{}_2$ on that side. The turning theorem therefore gives

$$\oint_{\partial D} \omega^1{}_2 + \sum_i (\pi - \alpha_i) = 2\pi.$$

Stokes gives $\oint_{\partial D} \omega^1{}_2 = \int_D K \mathrm{d}A$. Rearranging proves the formula. For this piecewise smooth boundary, Stokes follows by rounding each corner in a coordinate ball of radius ϵ. Smoothness bounds the changed connection integral by $O(\epsilon)$ and the changed curvature integral by $O(\epsilon^2)$. The replacement arcs have length $O(\epsilon)$, including at concave corners. Apply smooth-boundary Stokes to the rounded disk and take $\epsilon \to 0$. ∎

For example, the unit-sphere octant has curvature integral

$$\int_0^{\pi/2} \int_0^{\pi/2} \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\phi = \frac{\pi}{2}.$$

Its three right angles give the same excess $3\pi/2 - \pi = \pi/2$. This triangle is contained in a framed chart obtained by removing a point outside the octant from the sphere.

18.3 The local Gauss–Bonnet formula

Let $T = \dot\gamma$ be the unit tangent of a smooth boundary segment parametrized by arc length s. Its signed *geodesic curvature* is

$$k_g = g(\nabla_T T, JT).$$

For a positive boundary, JT points into the disk. Differentiating $g(T,T) = 1$ makes its covariant acceleration perpendicular to T. Writing $T = \cos\zeta\, e_1 + \sin\zeta\, e_2$ in the angle derivative already computed gives

$$\nabla_T T = (\zeta' - \omega^1{}_2(T))JT, \qquad k_g = \zeta' - \omega^1{}_2(T).$$

Thus geodesic sides have zero geodesic curvature. The corner diagram distinguishes a jump in the tangent from curvature on a smooth side.

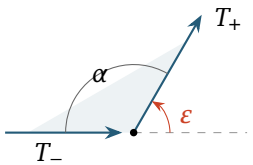


Figure 18.3: The exterior turn ε takes the incoming tangent to the outgoing tangent. The interior angle is $\alpha = \pi - \varepsilon$. A concave corner has negative exterior turn.

Theorem 18.3 (Local Gauss–Bonnet · *proved given turning and partitions of unity*)**.** *Let D be a compact embedded disk whose positive boundary has finitely many regular smooth sides and nondegenerate corners. Suppose one positive orthonormal frame is defined on a neighbourhood of all of D. Then*

$$\int_D K\,\mathrm{d}A + \oint_{\partial D} k_g\,\mathrm{d}s + \sum_i \varepsilon_i = 2\pi.$$

The side integrals exclude the corners; their turns occur in the finite sum.

Proof. Integrate $k_g = \zeta' - \omega^1{}_2(T)$ on each side. The turning theorem gives $\sum \Delta\zeta + \sum_i \varepsilon_i = 2\pi$. Summing the side integrals therefore gives

$$\oint_{\partial D} k_g\,\mathrm{d}s = 2\pi - \sum_i \varepsilon_i - \oint_{\partial D} \omega^1{}_2.$$

Use Stokes, with the same corner-rounding limit as above, to replace the last integral by $\int_D K\mathrm{d}A$. ∎

Consider a northern unit-sphere cap $0 \le \theta \le a$, where $0 < a < \pi$. Its positive boundary runs in increasing ϕ. Use the unit spherical frame $e_\theta = \partial_\theta$ and $e_\phi = (\sin\theta)^{-1}\partial_\phi$. The unit tangent is $T = e_\phi$ and $JT = -e_\theta$. Lecture 16's symbols give $\nabla_T T = -\cot a\, e_\theta$, so $k_g = \cot a$. The two integrals are

$$\int_D K\mathrm{d}A = 2\pi(1 - \cos a), \qquad \oint_{\partial D} k_g\,\mathrm{d}s = \cot a\,(2\pi \sin a) = 2\pi\cos a.$$

Their sum is 2π, with no corner contribution. A smooth frame on this cap is exhibited in the latitude example below.

18.4 Gauss–Bonnet on a closed surface

Theorem 18.4 (Gauss–Bonnet · *proved given cited topology and Stokes inputs*)**.** *Every closed oriented Riemannian surface satisfies*

$$\int_M K\mathrm{d}A = 2\pi\chi(M).$$

Proof. Use the cited existence of a finite smooth triangulation of a compact surface (Lee, discussion before Theorem 9.7, pp. 166–167). In Lee's definition, each curved triangle and its enclosed region lie in one coordinate chart, with regular sides and no cusps (p. 162). We also use the cited planar *Schoenflies theorem*: a simple closed plane curve and its enclosed region form a closed topological disk (Stillwell, §0.3.9, pp. 35–36). Gram–Schmidt on the connected chart neighbourhood of the face gives a positive orthonormal frame, changing one vector's sign if needed. Thus each closed face is a disk to which the local formula applies. The edges need not be geodesics.

To obtain the required Δ-complex, order the vertices and choose one parametrization for each edge, from its lower to its higher vertex. These give a homeomorphism from the boundary of each standard triangle to its face boundary. Identify the standard triangle with a closed unit disk by radially rescaling rays from its centroid. Schoenflies supplies a disk homeomorphism; it can be chosen to match that boundary map. Indeed, any difference

on the boundary circle is a homeomorphism $\psi\colon S^1 \to S^1$, which extends to the disk by $r\zeta \mapsto r\psi(\zeta)$, with the centre fixed. Composing with this extension matches the chosen boundary map. The face maps now agree on shared edges and give a finite Δ-complex structure on M.

Let V, E and F count vertices, edges and faces. Orient each face by the surface orientation and sum the local formula. Every shared edge has opposite boundary directions in its two faces. Reversing T leaves $\nabla_T T$ unchanged and negates JT, so it negates k_g. The two edge integrals cancel. At each vertex, the incident interior angles sum to 2π. Thus the sum of exterior turns over all face corners is $3\pi F - 2\pi V$. The figure displays these cancellations and the complete angle around a vertex.

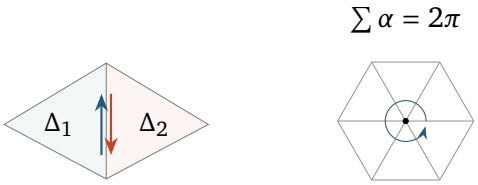


Figure 18.4: Adjacent faces traverse a shared edge in opposite directions. The incident face angles at an interior vertex fill one full turn.

The summed formula gives

$$\int_M K \mathrm{d}A = 2\pi F - (3\pi F - 2\pi V) = 2\pi V - \pi F = 2\pi(V - E + F).$$

The last equality uses $3F = 2E$, since each triangular side belongs to two faces on a surface without boundary. By Theorems 6.2 and 6.3, $V - E + F = \chi(M)$. For disconnected M, apply the argument to each component and add. ∎

The integral does not change when the metric changes, although the pointwise Gaussian curvature and the area form both change. A sphere therefore has total curvature 4π for every Riemannian metric. A torus has total curvature zero for every Riemannian metric, even when its curvature is not identically zero.

18.5 The Euler number of the tangent sphere

Recall from § 14.5 that the index of an isolated tangent-field zero is the winding of its normalized components on a small positive circle. The cited Poincaré–Hopf theorem, Theorem 14.5, states that the sum of these indices on a closed oriented surface is its Euler characteristic. In particular this sum is independent of the chosen field with isolated zeros. For the tangent bundle of S^2, call this common sum its *Euler number*.

> **Corollary 18.5** (The tangent-sphere Euler number · *sketched given Poincaré–Hopf*)**.** *The Euler number of TS^2 is $\chi(S^2) = 2$.*

The field $X(p) = e_3 - p_3 p$ from Lecture 14 has its two zeros at the poles, with index +1 at each. It supplies a field to which the definition applies. Poincaré–Hopf makes every such index sum equal to 2. A nowhere-zero tangent field would have empty index sum zero, contradicting this value. Hence there can be no global tangent frame on the sphere. A disk in one coordinate chart has a framed neighbourhood by Gram–Schmidt. The local curvature formula requires that the chosen frame extend over that disk, rather than only around its

boundary; the whole sphere admits no such frame. Lecture 22 expresses the same integer using changes between local frames.

18.6 Latitude holonomy and the Foucault pendulum

The *solid angle* of a region on the unit sphere is its area. Let the latitude be $\lambda \in (-\pi/2, \pi/2)$ and put $a = \pi/2 - \lambda$. Follow its circle once in increasing ϕ. The figure shows the northern cap and unit vectors obtained from the parallel equation.

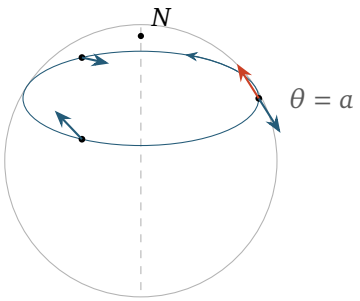


Figure 18.5: A latitude bounds the northern cap. At colatitude $a = \pi/3$, the arrows are projected values of the exact parallel field with frame angle $\alpha(\phi) = -\phi \cos a$. The red-orange arrow shows the final vector at the starting point after one full circuit.

Example 18.6 (Latitude rotation)**.** On the loop, the spherical frame has $\omega^1{}_2 = -\cos a \, \mathrm{d}\phi$. The parallel-angle equation gives

$$\Delta\alpha = -2\pi \cos a = -2\pi \sin \lambda.$$

The cap's solid angle is $\Omega_D = 2\pi(1 - \cos a) = 2\pi(1 - \sin \lambda)$. These real numbers differ by 2π and represent the same holonomy. The spherical frame itself is undefined at the north pole, so Stokes cannot be applied to that frame across the cap.

To use a frame on the full cap, define the rotated vectors by

$$e_{N,1} = \cos\phi \, e_\theta - \sin\phi \, e_\phi, \qquad e_{N,2} = \sin\phi \, e_\theta + \cos\phi \, e_\phi.$$

In Cartesian coordinates $p = (x, y, z)$ on the unit sphere, its columns are

$$E_N = \begin{pmatrix} 1 - x^2/(1+z) & -xy/(1+z) \\ -xy/(1+z) & 1 - y^2/(1+z) \\ -x & -y \end{pmatrix}.$$

The rotation makes it tangent, positive and orthonormal away from the poles. The displayed expression is smooth for $z > -1$ and at the north pole gives e_x and e_y. Continuity therefore preserves those properties there. Differentiating the rotated frame gives

$$\omega_N = \omega^1{}_2 + \mathrm{d}\phi = (1 - \cos\theta)\mathrm{d}\phi = \frac{x\mathrm{d}y - y\mathrm{d}x}{1+z}.$$

This is smooth at the north pole. Its integral around the latitude is $2\pi(1 - \cos a)$, so Stokes in this frame gives the cap's curvature integral. At the equator the holonomy is zero modulo 2π, while the cap integral is 2π. Holonomy alone does not retain that full turn.

For the physical interpretation, use the small-swing Coriolis model of an ideal pendulum on a slowly rotating sphere.[1] Let $\nu_\oplus > 0$ be the sphere's rotation frequency and $\nu_0 > 0$ the pendulum's natural frequency, with $\nu_\oplus \ll \nu_0$. In a positive east–north frame, write the small displacement as $\xi = \xi_E + \mathrm{i}\xi_N$. Define the local Coriolis rate $b = \nu_\oplus \sin\lambda$. Keeping the linear Coriolis term gives

$$\ddot{\xi} + 2\mathrm{i}b\dot{\xi} + \nu_0^2\xi = 0.$$

Substituting $\xi = e^{-\mathrm{i}bt}w$ cancels the first-derivative terms and gives

$$\ddot{w} + (\nu_0^2 + b^2)w = 0.$$

An oscillation along a fixed line in the w plane therefore has slow orientation angle $-bt$. Its rate agrees with parallel transport:

$$\dot{\alpha} = -\cos a\,\dot{\phi} = -\nu_\oplus \sin\lambda = -b.$$

Over one rotation $2\pi/\nu_\oplus$, the change is $-2\pi\sin\lambda$. It is clockwise in the northern hemisphere and counter-clockwise in the southern. An oriented swing direction records angles modulo 2π; the physical unoriented swing line records them modulo π. For a spin-1/2 ground state aligned with a field direction traversing this latitude, Lecture 24 derives the Berry phase $-\Omega_D/2$ modulo 2π.

Self-check. Check global Gauss–Bonnet for the round sphere and the flat quotient torus.

Answer. On a sphere of radius R, curvature $1/R^2$ times area $4\pi R^2$ gives $4\pi = 2\pi\chi(S^2)$. On the flat torus $K = 0$, so the integral is $0 = 2\pi\chi(T^2)$, using $\chi(T^2) = 0$ from Lecture 6.

Summary. Surface transport has angle $\int_D K\mathrm{d}A$ modulo 2π when one positive orthonormal frame covers the disk. The turning theorem adds the boundary tangent and corner turns, giving local Gauss–Bonnet. Using the cited topology inputs and summing over the triangulation yields $\int_M K\mathrm{d}A = 2\pi\chi(M)$ on every closed oriented Riemannian surface. The tangent sphere has Euler number 2 by the cited Poincaré–Hopf theorem. Latitude transport gives the ideal Foucault angle $-2\pi\sin\lambda$ modulo 2π. Lecture 19 uses curvature and geodesics in a Lorentzian spacetime.

Reading. Tu, *Differential Geometry*, §17, especially Theorems 17.4–17.5 and 17.9, pp. 142–146. Lee, *Riemannian Manifolds: An Introduction to Curvature*, Ch. 9, pp. 158–168, including the smooth triangulation discussion. Stillwell, *Classical Topology and Combinatorial Group Theory*, §0.3.9, pp. 35–36 (Schoenflies theorem).

[1] See Fitzpatrick, *Newtonian Dynamics*, Foucault Pendulum, for the linear Coriolis model, and Dray, *The Foucault Pendulum* (1998), for its slowly changing direction and tangential-projection interpretation.